

4.1 Simple Phenomena of magnetism

Properties of magnetism

- Forces are strongest at north and south poles
 - When 2 magnets are held together, there will be a force between the magnets
 - Magnetic force is due to interactions between magnetic fields
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- Unlike poles attract, because their magnetic field lines merge, creating a low energy state that pulls the magnets together
 - Like poles repel, because their magnetic field lines clash, creating a high energy state that pushes the magnets apart
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Magnetic materials

- Materials that are attracted to magnets (regardless of pole)
- Materials: iron, nickel, cobalt and steel

Hard magnetic materials:

- Materials that are difficult to magnetise, and retains magnetism well once magnetised
- Example: steel
- Uses: Permanent magnets, compass needles, loudspeaker magnets, fridge magnets

Soft magnetic materials:

- Materials that are easy to magnetise and demagnetise
- Example: soft iron
- Uses: Cores for electromagnets, Radio aerials, Transformers

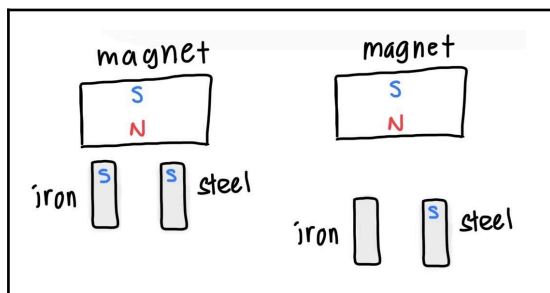
Non-magnetic materials

- Materials that are not attracted to magnets
 - Materials: all metals except iron, nickel and cobalt, and all non-metals
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How to determine whether a material is a magnet, magnetic material, or non-magnetic material:

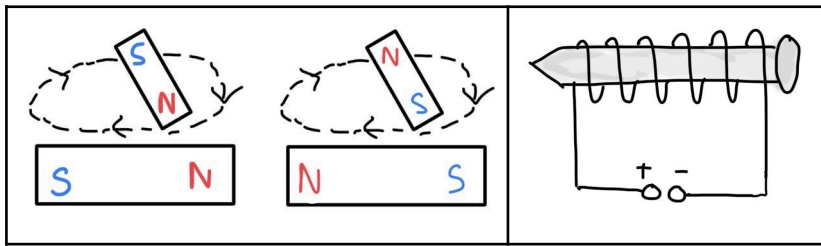
- Bring it close to a known magnet
- If material can be attracted and repelled, it is a magnet
- If material can only be attracted, it is a magnetic material
- If material is not affected, it is a non-magnetic material

Induced magnetism of magnetic materials by a strong magnet:



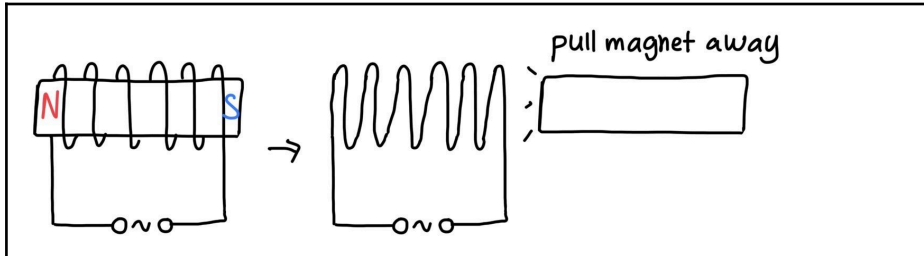
- When an iron bar and steel bar is brought close to a magnet, the iron and steel bar are magnetised
 - Poles are induced in the iron and steel bar
 - When the bars are brought away from the magnet, the iron bar loses its magnetism, while the steel bar is permanently magnetised
-

Magnetising a magnetic material:



- Stroke material repeatedly in one direction using a magnet
- Place material in a coil carrying direct current (DC)
 - Insert the material a long coil of wire
 - DC Power supply is turned on, to pass a large direct current through the coil
 - Current generates a magnetic field that magnetises the material

Demagnetising a magnet:



- Heat magnet to a high temperature
- Hit the magnet repeatedly using a hammer
- Place magnet in a coil carrying alternating current (AC)
 - Insert the magnet into a long coil of wire
 - Turn AC power supply on, to pass a large alternating current through the coil
 - Slowly pull the magnet away from the coil, until it is completely outside the coil.
 - The alternating current generates a changing magnetic field to demagnetise the magnet

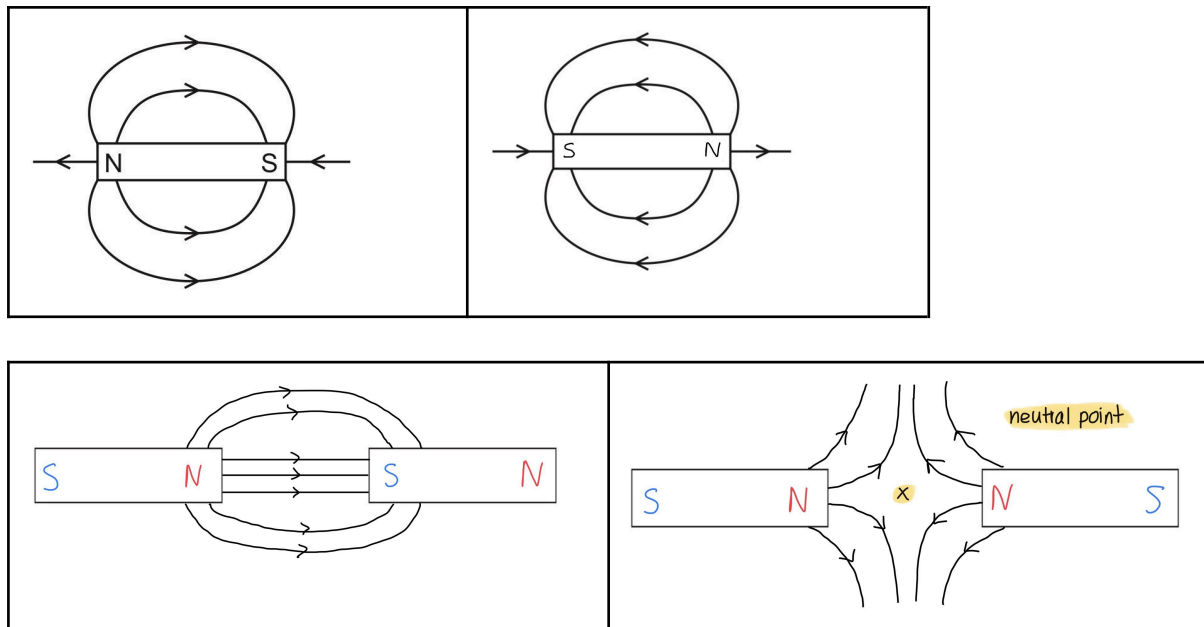
Note: has to be until magnet is completely outside of the wire before turning off power supply, or else it will not be demagnetised

Magnetic field: The region in which a magnetic pole experiences a force

Direction of a magnetic field at a point: The direction of the force on the North Pole of a magnet at that point

- Magnetic field lines are lines which represent the direction and strength of a magnetic field
 - They go from North to South Pole
 - There is strong magnetic field strength where lines are closer together
 - They cross each other

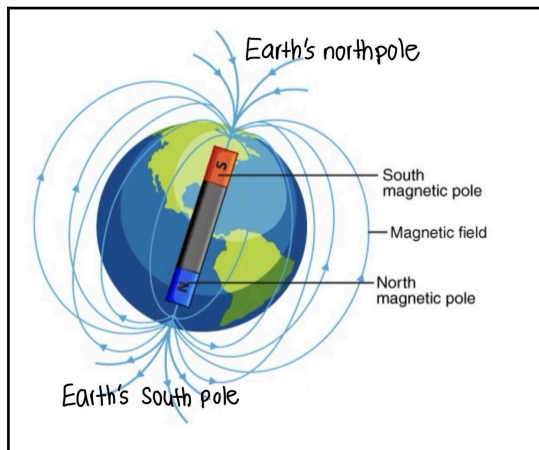
Diagrams of magnetic field lines around bar magnets:



Uniform magnetic field:

- Field lines are parallel and same distance from each other
- This shows that the magnetic field strength is constant

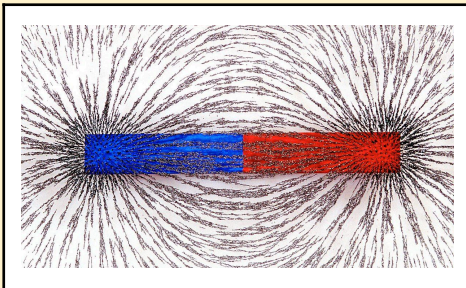
Magnetic field of the Earth:



- Earth's Geographic North = Magnetic South
 - Needle of compass is always pointed to the magnetic south, which is the geographic north
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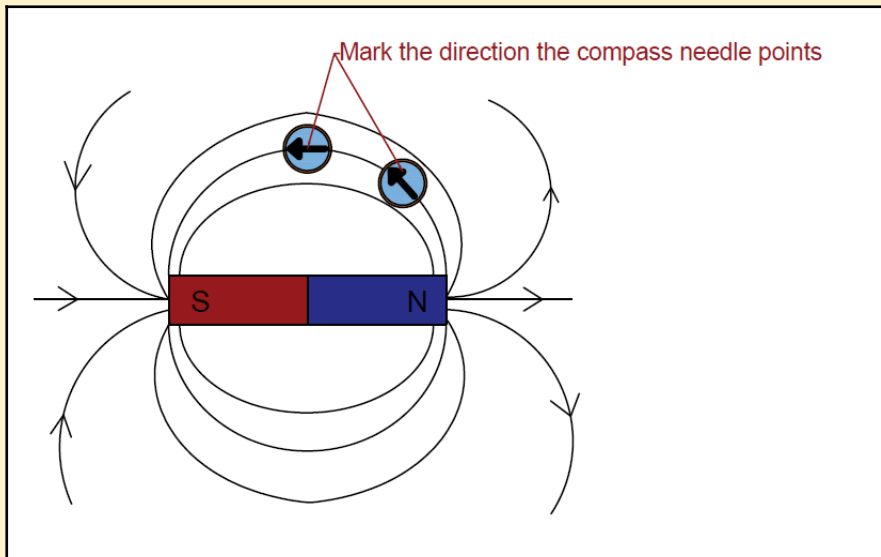
Plotting magnetic field lines around a bar magnet

Using iron fillings:



- Place a piece of paper on top of the magnet
- Gently sprinkle iron fillings on top of the paper
- Carefully tap the paper to allow the iron fillings to settle onto the magnetic field lines

Using a compass:



- Place magnet on top of a piece of paper
- Place the compass at one end of the magnet. Using a pencil, mark a dot at the tip of the needle
- Move the compass so that the end of the needle lines up with the previous dot, and then mark a dot at the tip of the needle again
- Repeat until there is a chain of dots, then link the dots in a smooth curve. This is a magnetic field line
- Repeat process from another position next to the magnet

- The tip of the needle is a North Pole, which points to the magnet's South Pole
 - The end of the needle is a South Pole, which points to the magnet's North Pole
-

4.2 Electrical Quantities

Electric charge:

- Objects can be given a positive or negative charge
- When 2 charged objects are brought together, there is a force between the objects
 - Like charges repel
 - Unlike charges attract
- Electric charge is measured in Coulombs (C)

Where does charge come from?

- Everything is made out of atoms. There are electric charges inside them.
 - Protons -> positive
 - Electrons -> negative
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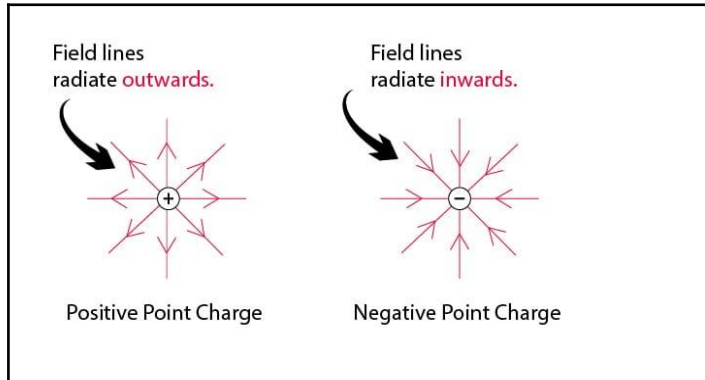
Electric field: the region in which an electric charge experiences a force

Direction of an electric field at a point: the direction of the force on a positive charge at that point

- Electric field lines are lines which represent the direction and strength of an electric field
 - They go from positive to negative charges
 - There is strong electric field strength where lines are closer together
 - They cannot cross each other
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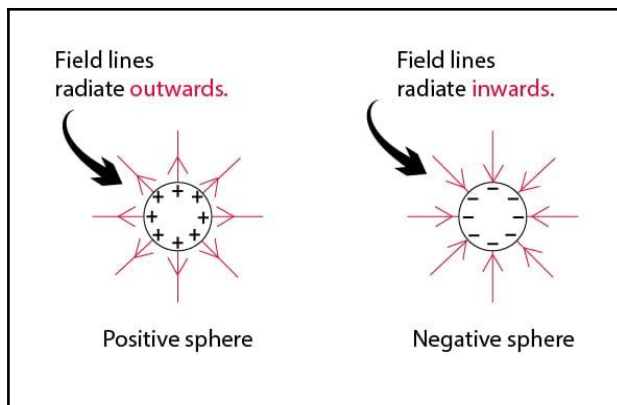
Field lines around a point charge:

- Field lines are symmetrical
- Field lines radiate outwards or inwards



Field lines around a charged sphere:

- Field lines are symmetrical, because charges on the surface of the sphere are evenly distributed
- Field lines radiate outwards or inwards



Field lines between 2 oppositely charged plates:

- Uniform electric field - field lines are parallel and same distance from each other
 - This shows that electric field strength is constant
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Conductors and insulators

Conductors:

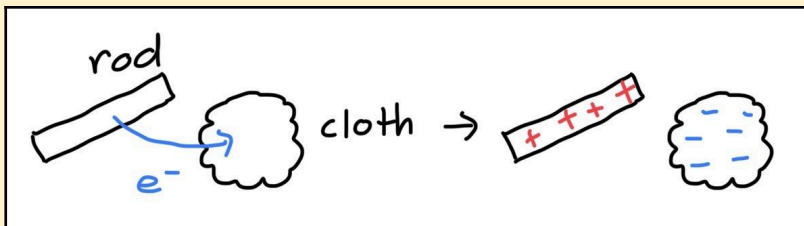
- Materials that allow charge to flow through it easily, because it has free moving electrons to carry charge
- Example - iron, copper, graphite

Insulators:

- Materials that does not allow charge to flow through it easily, because it has no free moving electrons to carry charge
- Example - rubber, plastic, glass

*Note - the electrons in insulators are held tightly in atom, which is why they are unable to move (not free moving)

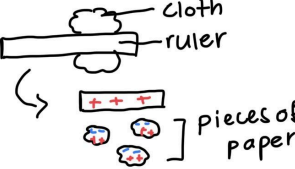
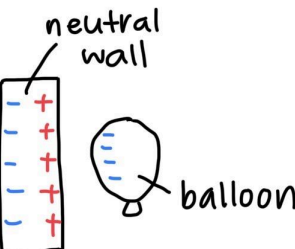
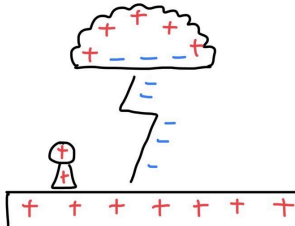
Charging by friction: (with 2 insulators)



- When an uncharged plastic rod is rubbed with an uncharged cloth, friction causes electrons to transfer from the rod to the cloth
- The rod loses electrons, it becomes positively charged
- The cloth gains electrons, it becomes negatively charged
- They are oppositely charged, so they attract

Note: connecting wire is made out of copper, because copper is a good conductor of electricity

Electrostatic phenomena

 A hand-drawn diagram showing a comb with a wavy line representing hair being attracted to it. Labels: 'Comb' and 'Hair'.	- The comb attracts hair when brushing. - This is because friction causes electrons to be transferred, and comb and hair to be charged - They become oppositely charged, so they attract
 A hand-drawn diagram showing a ruler with a cloud-like shape above it. Below the ruler are several small pieces of paper, some with '+' signs. Labels: 'cloth ruler' and 'pieces of paper'.	- When the positively charged ruler is held close to small uncharged pieces of paper, the electrons within the paper will be attracted to the ruler
 A hand-drawn diagram showing a balloon with '-' signs and a stream of water bending towards it. Labels: 'balloon' and 'stream of water'.	- The negatively charged balloon attracts the positive charges in the stream of water, causing it to bend towards the balloon
 A hand-drawn diagram showing a balloon with '-' signs and a wall with '+' signs. Labels: 'neutral wall' and 'balloon'.	- When a negatively charged balloon strikes a wall, it repels the negative charges in the atoms of the wall, causing them to move away from the balloon - The surface of the wall is now positively charged, so the negatively charged balloon is attracted to it and sticks to it
 A hand-drawn diagram showing a cloud with '+' signs and a lightning bolt striking the ground, which has '+' signs. Labels: 'cloud' and 'ground'.	- Negative charges at the bottom of the cloud travel to the ground, because they are attracted to the positive charges on the ground - This creates an electric current

- The reason why uncharged objects are attracted instead of repelled, is because the attractive forces between unlike charges are greater than the repulsive forces between like charges.
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Current

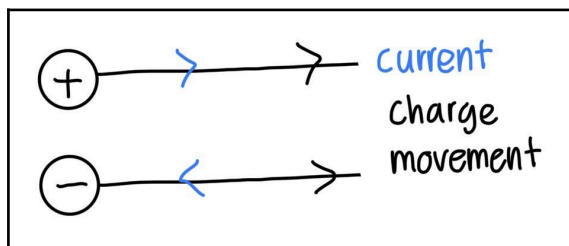
Electric current: the charge passing a point per unit time

- Electric current is the rate of flow of charge

$$I = \frac{Q}{t}$$

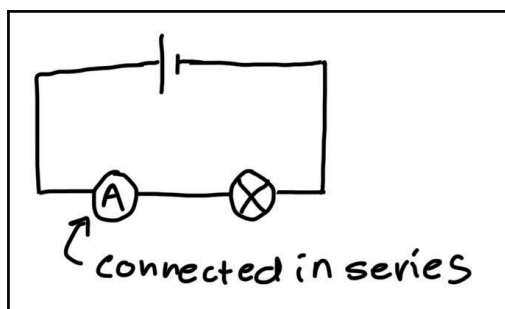
current (A) charge (C) time (s)

Direction of current:



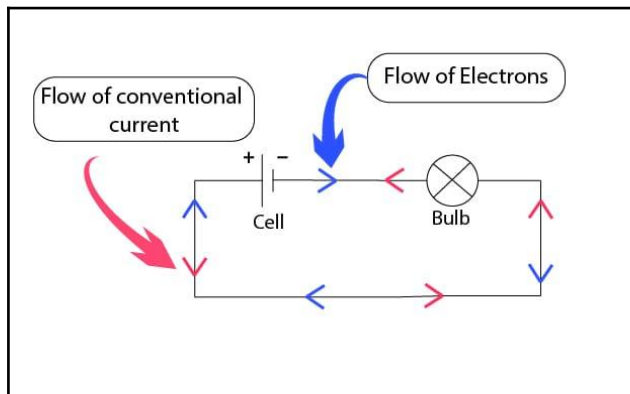
Measuring current:

- Ammeter is used to measure current in a circuit
- It must be connected in series
- A good ammeter has a low resistance, to allow more charges to flow through it



Note - connecting wire is made out of copper, because copper is a good conductor

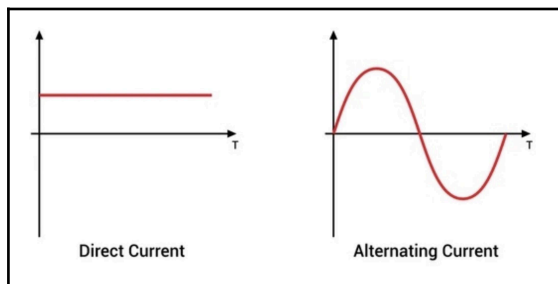
Current and free moving electrons



- Conventional current - flows from positive to negative terminal
- Free moving electrons - flows from negative to positive terminal

Direct current: current that flows in one direction only

Alternating current: current that flows forwards and backwards



Electromotive force and potential difference

Electromotive force (emf): the electrical work done by a source in moving a unit charge around a complete circuit

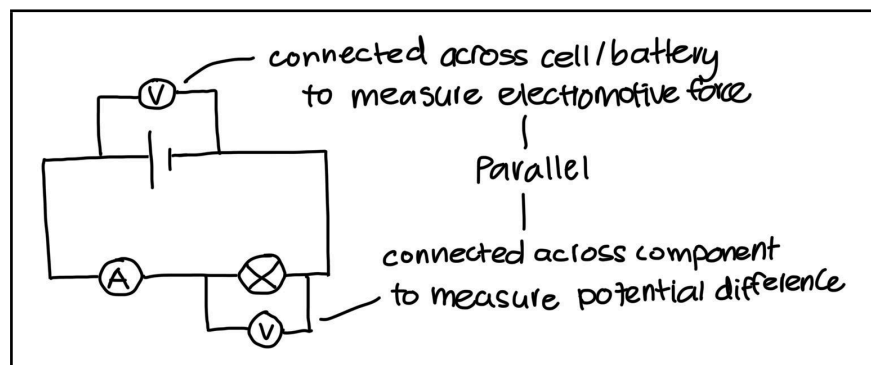
Potential difference (pd): the work done by a unit charge passing through a component

$$\begin{array}{l} \text{emf (V)} \end{array} E = \frac{W}{Q} \begin{array}{l} \text{work (J)} \\ \text{charge (C)} \end{array}$$

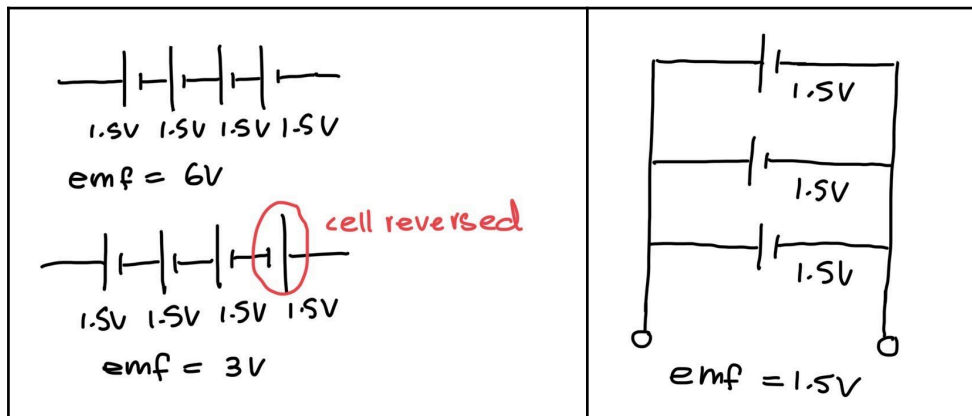
$$\begin{array}{l} \text{pd (V)} \end{array} V = \frac{W}{Q} \begin{array}{l} \text{work (J)} \\ \text{charge (C)} \end{array}$$

Measuring electromotive force and potential difference:

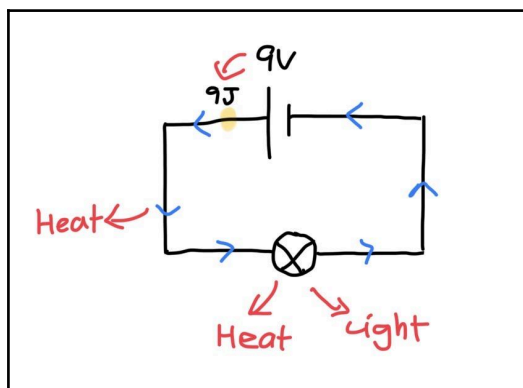
- Voltmeter is used to measure electromotive force of battery/cells, and potential difference across components
- It is connected in parallel with the battery/cell or component
- A good voltmeter has a high resistance to block current flowing through it



Connecting cells in series and in parallel:



How the Circuit works



- Cells/batteries move charges around circuits
 - They transfer energy to the charges
 - This energy is measured in volts (Joules per coulomb)
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- For a 9 volts cell, 9 joules of energy is given to each coulomb of charge that passes through the cell
 - As charges flow around a circuit, the energy they carry is transferred to the components they pass through
-
- 1) When current passes through the wires, energy is transferred to the surroundings as heat
 - 2) When current passes through a bulb, energy is transferred to the surroundings as heat and light

Resistance

Resistance: the ratio of potential difference to current

$$R = \frac{V}{I}$$

resistance (Ω) voltage (V)
current (A)

- Resistance is the value in the material that resists the current to flow through it

Link to Ohm's law:

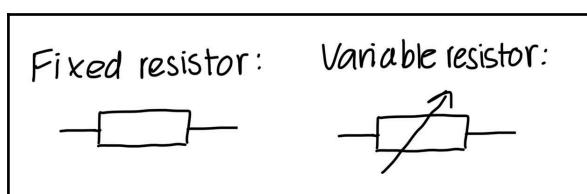
- Potential difference across a metallic conductor is directly proportional to the current passing through it, when temperature is constant

$$V \propto I \rightarrow V = IR$$

- When a metallic conductor obeys Ohm's law, it is called an ohmic conductor

Resistor

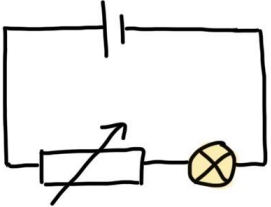
- The component that controls current
- If the circuit has no resistor, the current may be too high, and the bulb will break



- Variable resistor can vary resistance, can be used to vary the current in a circuit

Resistance and resistor in circuits:

- For resistor, when charge passes through it, electrical energy is transferred to heat, which is lost to surroundings

	- If resistance is increased, current will be smaller and bulb will glow less brightly - If resistance is decreased, current will be larger and bulb will glow more brightly
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Factors that affect the resistance of a metallic conductor

Temperature:

- As temperature increases, resistance increases
- Atoms gain kinetic energy and vibrate more, which reduces the rate of flow of charge -> current decreases, meaning resistance increases

Type of material:

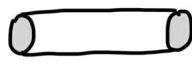
- Different materials have different resistance
 - Conductors have a lower resistance
 - Insulators have a higher resistance
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Factors that effect the resistance of a wire

Length of a wire:

$L \uparrow$ $R \uparrow$  $L \downarrow$ $R \downarrow$ 	- The longer the wire, the greater the resistance - Electrons have to collide with more metal ions, reducing rate of flow of charge - Current decreases, meaning resistance increases
$R \propto L$	So, resistance is directly proportional to length

Cross section area of a wire:

$A \uparrow$ $R \downarrow$  $A \downarrow$ $R \uparrow$ 	- The thicker the wire, the smaller the resistance - Electrons have more space to flow, increasing rate of flow of charge - Current increases, meaning resistance decreases
$R \propto \frac{1}{A}$	So, resistance is inversely proportional to cross-section area
$R \propto \frac{1}{r^2} \propto \frac{1}{d^2}$	

Note -> more thicker, meaning larger cross-section area

Example 1:

$$\begin{array}{ll}
 R_1 = 0.45 \Omega & \text{For length,} \\
 \text{length} = l & R \propto l = \frac{1}{3} \\
 \text{area} = A & \text{For area,} \\
 R_2 = ??? & R \propto \frac{1}{A} = 2 \\
 \text{length} = \frac{1}{3}l & \\
 \text{area} = \frac{1}{9}A &
 \end{array}$$

multiplier: $2 \times \frac{1}{3} = \frac{2}{3}$

$$R_2 = 0.45 \times \frac{2}{3} = 0.30 \Omega \checkmark$$

Example 2:

Wire P:

$$R = 330 \Omega$$

diameter = d

length = L

wire Q:

$$\text{diameter} = \frac{1}{2}d$$

$$\text{length} = 5L$$

For diameter,

$$R \propto \frac{1}{d^2} = (2)^2 = 4$$

For length

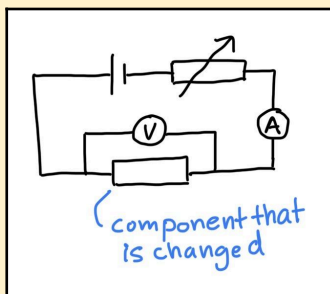
$$R \propto L = 5$$

$$\text{multiplier: } 5 \times 4 = 20$$

$$R(\text{wire Q}) = 330 \times 20 = 6600 \Omega \checkmark \#$$

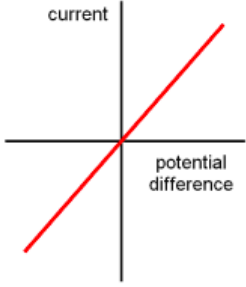
Current-voltage graphs

Experiment to obtain current-voltage graphs for a resistor, filament lamp, and diode



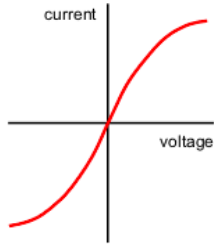
- Set up as shown in the diagram.
- Set up circuit with cell, variable resistor, (resistor), voltmeter and ammeter. Ammeter is connected in series and Voltmeter is connected in parallel.
- Alter the value of the variable resistor to 10Ω , then measure voltage using voltmeter and current using ammeter
- Repeat using 20Ω , 30Ω , 40Ω , and 50Ω
- Reverse the cell and repeat the steps
- Plot a graph of current against voltage

For resistor:

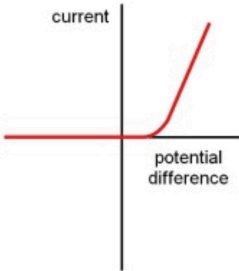
	- Straight line passing through the origin - Gradient is constant, so resistance is constant - Current is directly proportional to voltage
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For filament lamp:

- Please be aware if it's a current/voltage graph or voltage/current graph in the question -> descriptions will change

	- Graph is not straight line - Gradient is not constant, so resistance is not constant - When current increases, temperature increases so resistance increases, causing gradient to decrease - When current decreases, temperature decreases so resistance decreases, causing gradient to decrease
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For Diode:

	- Diodes have a high resistance when current is in one direction, and a low resistance when current is in the opposite direction - Explanation: diode allows current to flow in one direction only
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Heating effect of current by resistance

- Heating effect is in wires or appliances, and is caused by their resistance and current passing through them

For a house:

- The wiring in a house is designed to let current pass through it easily
- Wires do not become warm when appliances are being used
- Wires have a low resistance, and lower heating effect

For a light bulb:

	- When current passes through the very thin wire (filament) of a traditional light bulb, it becomes very hot and glows white. - For filament, when wire is thinner, electrons have less space to flow, decreasing current and increasing resistance - So filament has high resistance and electrical energy is transferred to heat and light energy
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Common appliances that make use of heating effect:

- Kettles, electric ovens, hair dryers
- Wires are the heating elements
- They want wires to be warm -> so wire is designed to have a high resistance, so that as current passes through them, energy is transferred to heat energy and element heats up

Electrical Power

$$P = IV$$

power (W) current (A) voltage (V)

$$P = I^2 R$$

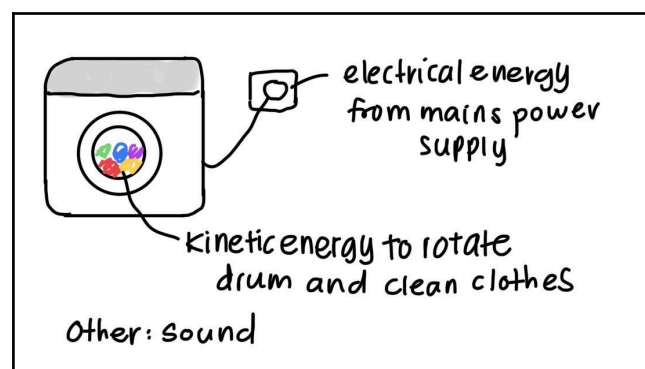
power (W) current (A) resistance (Ω)

Electrical Energy

$$E = VIt$$

energy (J) voltage (V) current (A) time (s)

- Appliances transfer electrical energy from mains electricity to other forms of energy
- The amount of energy an appliance transfers depends on:
 - How long the appliance is switched on
 - The power of the appliance



Measuring energy usage:

- Energy usage in homes and businesses is calculated using kilowatt-hour (kWh)
- This derived from the $E=Pt$ formula

Kilowatt-hour (kWh): energy transferred in one hour, at a rate of transfer of one kilowatt

- remember, 1 kilowatt = 1000 watts
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Topic 4: Magnetism & Electricity (4.3 - 4.4)

4.3 Electric circuits

Circuit diagrams and circuit components

Power supply, cell and battery:

- Supplies electrical energy to the circuit, which moves charges around the circuit -> causing the current

Lamp:

- Glows when current flows through it

Ammeter:

- Used to measure current in the circuit
- Connected in series

Voltmeter:

- Used to measure potential difference and emf in the circuit
- Connected in parallel

Switch:

- Used to turn on and off the circuit
- Open switch = circuit closes
- Closed switch = circuit opens

Fuse:

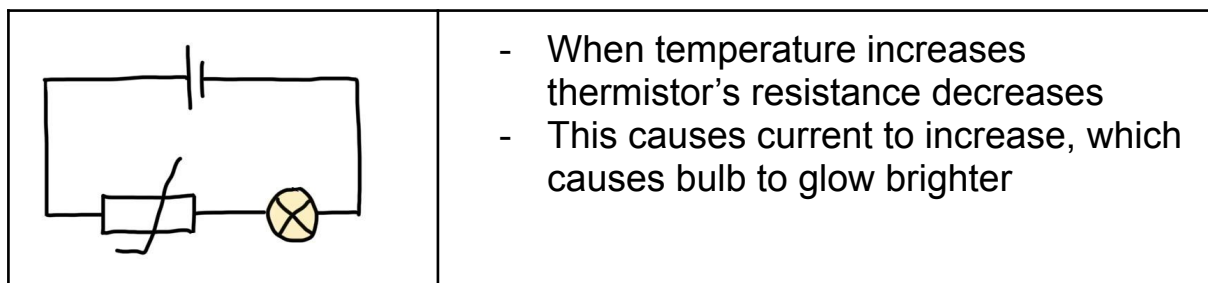
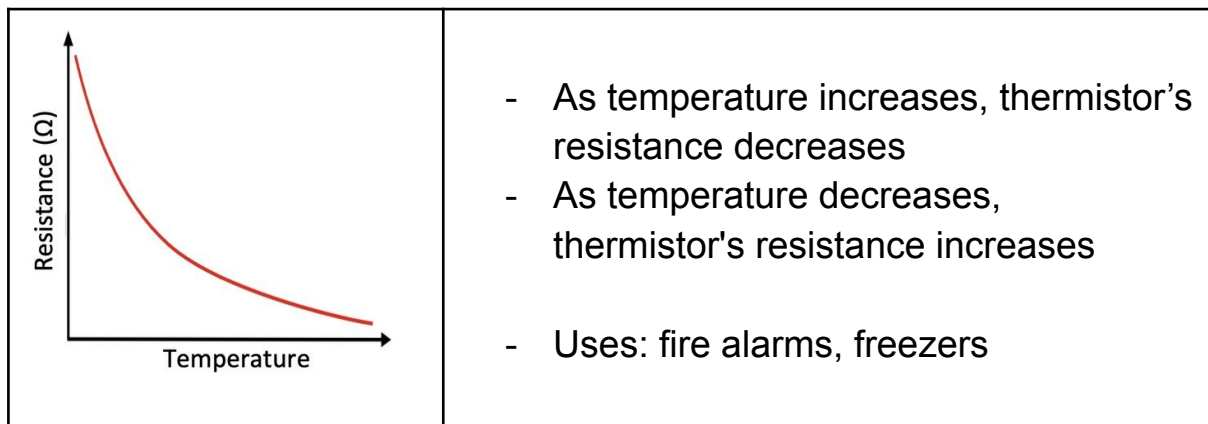
- Used to prevent excessive current in the circuit, and act as a safety against fire
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Fixed resistor and variable resistor:

- Used to control current
- The higher the resistance, the smaller the current

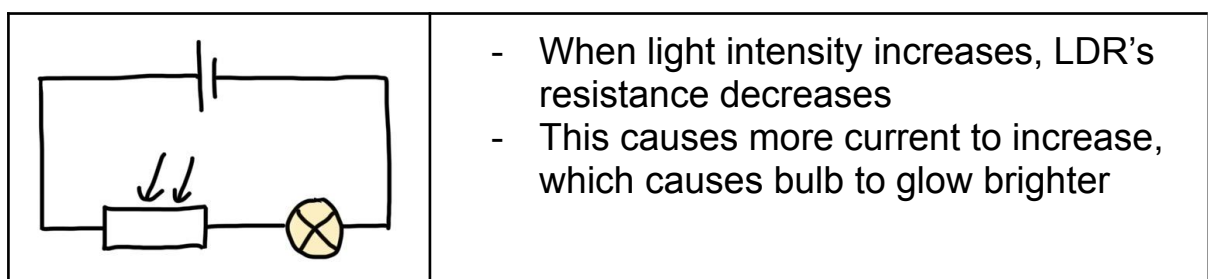
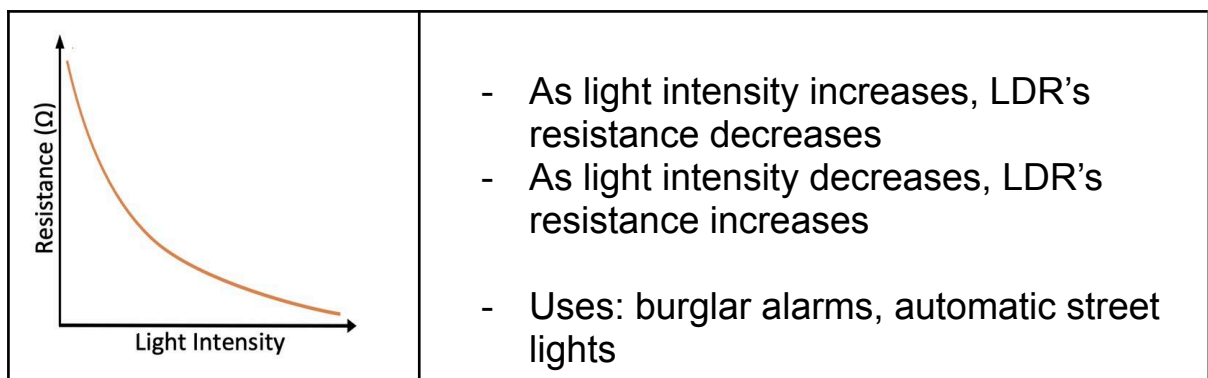
Thermistor:

- Temperature sensor



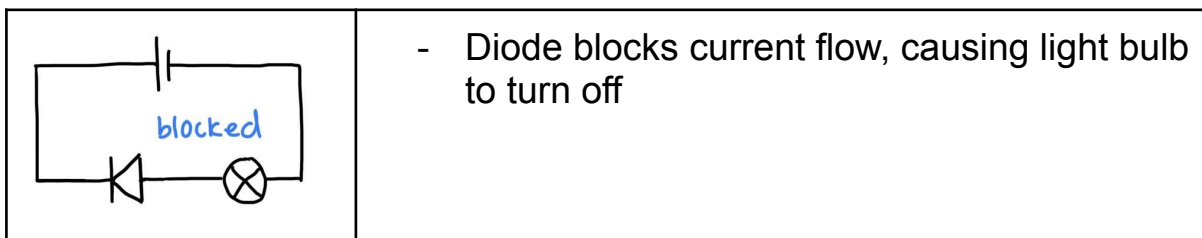
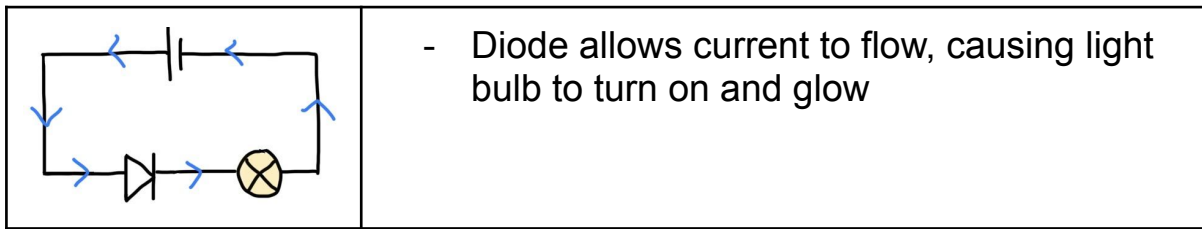
Light dependent resistor (LDR)

- Light sensor / detector



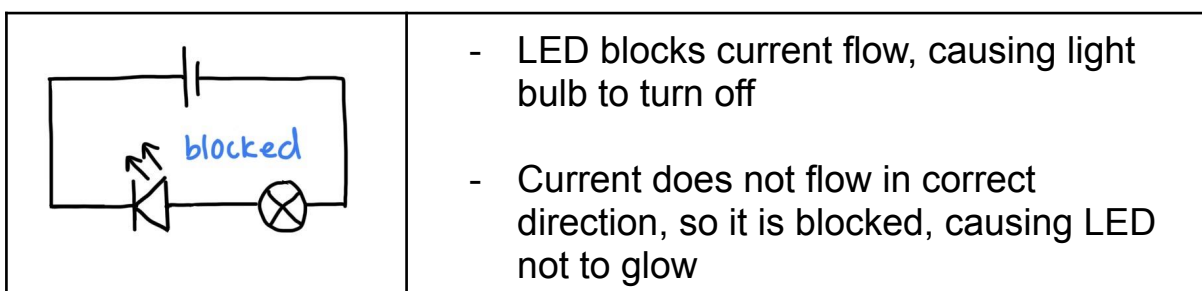
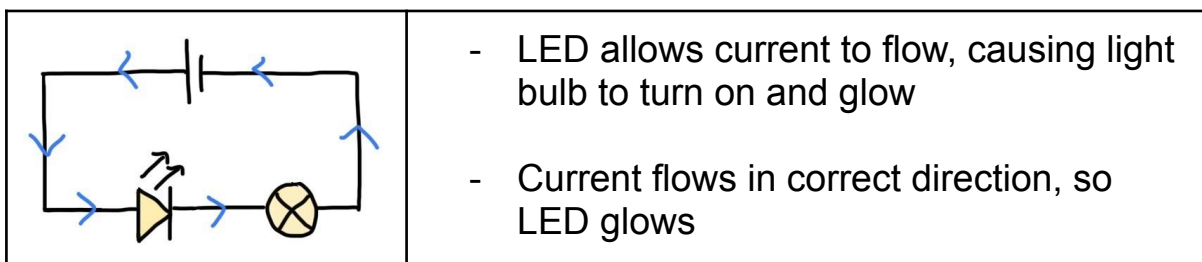
Diode:

- Allows current to flow in one direction only



Light emitting diode: (LED)

- Glows when current flows through in correct direction



- Diodes are used in rectifier circuits to convert alternating current into direct current
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Heater:

- Device that converts electrical energy into heat energy

Motor:

- Device that converts electrical energy into kinetic energy

Generator:

- Device that converts kinetic energy into electrical energy
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Relay coil:

- Used to control a large current using a small current

Transformer:

- Device that converts voltage from high to low, or low to high

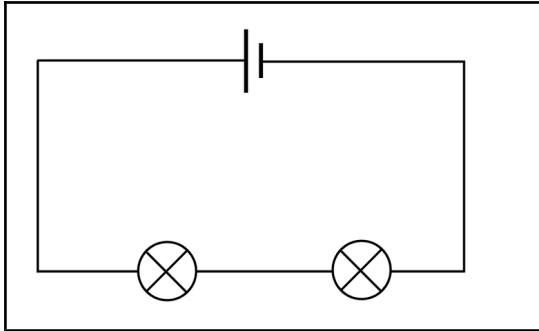
Potential divider or potentiometer:

- A kind of variable resistor that consists of a coil of wire with a sliding contact midway along it
 - The resistance can be varied by moving the sliding contact, which changes the length of the wires and the resistance of the potential divider
-

Series and Parallel circuits

Series circuit:

- A simple loop circuit with no branches, only one path for current to flow



- Current at every point in a series circuit is the same

$$I_T = I_1 = I_2 = I_3$$

- The total potential difference across the components is equal to the sum of the individual potential difference of each component

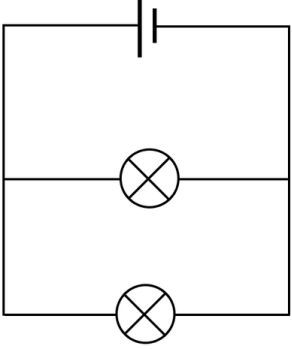
$$E \text{ or } V_T = V_1 + V_2 + V_3$$

- The total resistance of the components is equal to the sum of the individual resistance of each component

$$R_T = R_1 + R_2 + R_3$$

Parallel circuits:

- A circuit with branches, providing more than one path for current to flow

	- Switches can be placed on different parts of the circuit to switch each component on and off individually, or all together - If one component breaks, only components on the same branch are effected
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- The total current from the source is equal to the sum of the current of each branch

$$I_T = I_1 + I_2 + I_3$$

- The electromotive force is equal to the potential difference across each component

$$E \text{ or } V_T = V_1 = V_2 = V_3$$

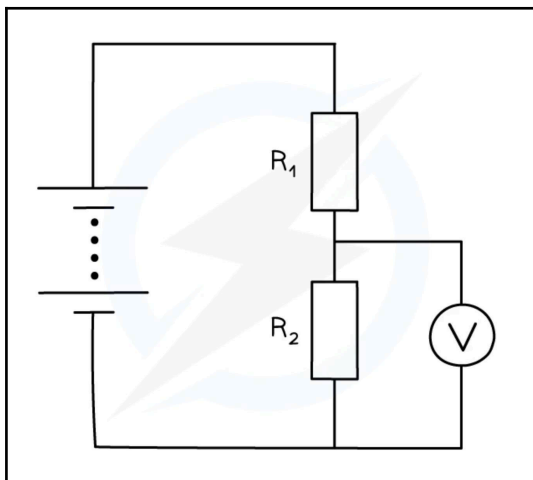
$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

Advantages of Parallel circuits over Series circuits:

- If one component breaks, only components on the same branch will stop working -> other components will still work
- Bulbs have equal brightness, as potential difference is equal
- Able to connect switch to turn each component on and off individually, or all components together

Potential dividers

Potential divider: divider that splits the electromotive force from a power source between 2 or more components in series



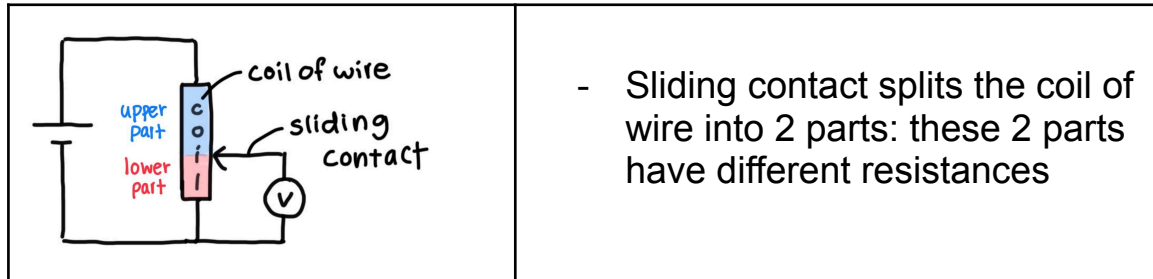
When 2 resistors are connected in series:

- The current remains the same at any point in a series circuit
 - The electromotive force is shared between them
 - The total resistance is shared between them
-
- The potential difference across each resistor is proportional to its resistance, and this leads to:

$\frac{R_1}{R_2} = \frac{V_1}{V_2}$	- This is like ratio, so they may give you the emf, and you will need to find V_1 and V_2 independently based on the ratio of the resistance
-------------------------------------	--

Potentiometer (rheostat)

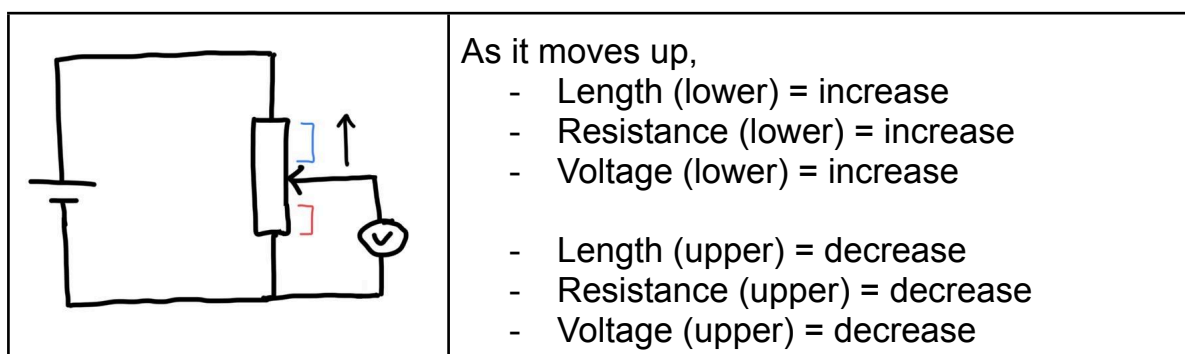
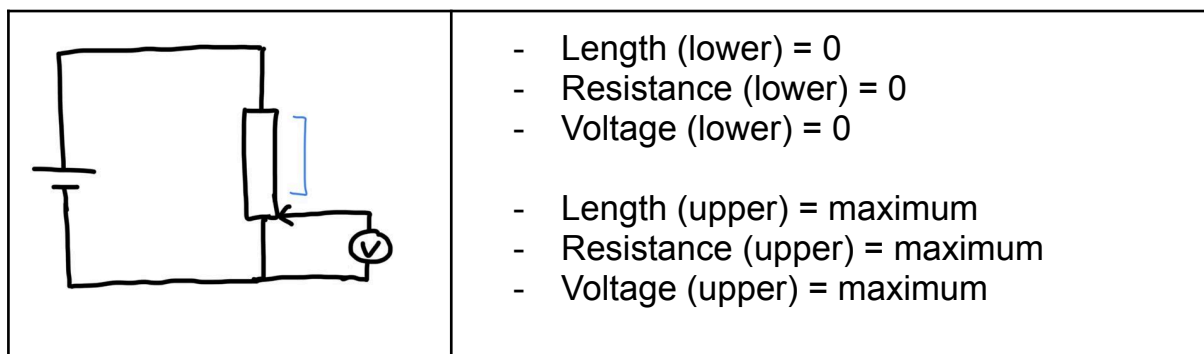
- Type of potential divider - consists of a coil of wire with a sliding contact midway along it



- Resistance is directly proportional to length
- Voltage is proportional to resistance
- So, voltage is directly proportional to length

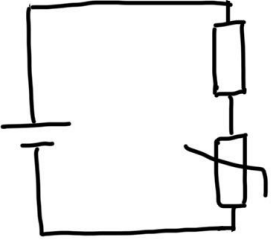
$R \propto L$ $V \propto L$ $\xrightarrow{\text{since } R \propto V}$

Sliding contact at different positions:

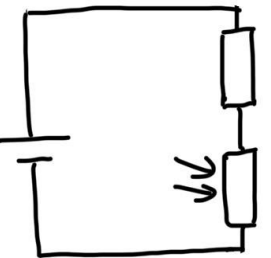


Potential divider and thermistor:

- Same as before (same formula applies)

	As temperature increases, - Resistance (thermistor) decreases - Voltage (thermistor) decreases - Resistance (resistor) increases - Voltage (resistor) increases
---	--

Potential divider and light-dependant resistor:

	As light intensity increases, - Resistance (LDR) decreases - Voltage (LDR) decreases - Resistance (resistor) increases - Voltage (resistor) increases
--	--

4.4 Electrical Safety

Electrical hazards from mains electricity

Damaged insulation:

- If someone touches an exposed piece of wire, they can get electrocuted

Overheating of cables:

- this can cause a fire, or melt insulation, exposing live wire and causing electrocution
- Caused by too much current through too small of a wire
- Caused by a long length of wire tightly coiled

Damp conditions:

- If moisture comes into contact with live wires, the moisture can conduct electricity, causing electrocution

Excess current from overloading plugs and sockets:

- When overloading (plugging in too many components), the heat created can cause fires
-

Mains Electricity

- It is an alternating current
- Mains circuit consists of 3 wires: live wire, neutral wire and earth wire

Live wire:

- Brown insulated
- Provides path for alternating current to travel, from power source to appliance

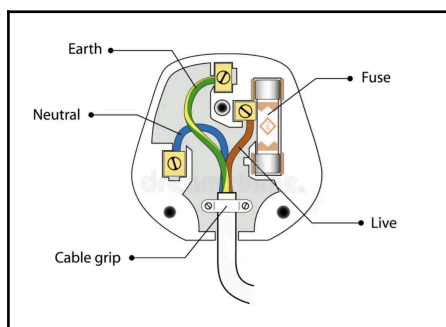
Neutral wire:

- Blue insulated
- Completes the circuit by providing a return path for current to travel back to power source

Earth wire:

- Yellow green insulated
- Protects the user if appliance is at fault
- Provides a path for current to escape without passing through the user, by flowing it directly to the ground

Plugs and sockets - 3 pin plug:



When wiring a plug:

- Ensure wires are connected to correct terminals
- Ensure cables are held firmly by the grip
- Ensure a fuse with an appropriate fuse value is fitted

Fuse:

- Has to be connected to the live wire
- When too large current flows through the live wire, the fuse will melt, breaking the circuit
 - This prevents electrocution and cables overheating, causing a fire
- Once the fault has been corrected, the fuse is blown
- It must be replaced with a new one of the same size, before the appliance can be used again

Circuit breakers: (such as trip switches)

- Also has to be connected to the live wire
- When too large current flows through the live wire, the switch automatically opens, making the circuit incomplete
- Once the fault is corrected, it does not need to be replaced.
- Instead, the switch is reset by pressing the reset button.

Switch:

- Connected in the live wire
- So, when the appliance is turned off and the switch is open, the current is completely disconnected from the appliance.
- The user is safe from being electrocuted
- When connected to the neutral wire, when the appliance is turned off and switch is open, the current is still connected to the appliance
- The user can get electrocuted

Earth wire:

- The Earth wire is the wire that provides a path for current to escape without passing through the user, by flowing it directly to the ground
 - This prevents the appliance from being live and causing electrocution
- If the live wire comes loose and touches the metal body, the current will immediately flow to the ground through the earth wire.
- The fuse will be blown or the circuit breaker will be tripped, breaking the circuit
- However, if appliance has no earth wire, when the live wire becomes loose and touches the metal body, The current can flow through the user, which causes electrocution.

Double Insulated appliance:

- Some appliances do not have an earth wire. This is because their outer case is made out of plastic
 - The plastic acts as a layer of insulation around the wires.
 - So, even when live wire becomes loose, the current cannot flow through the user -> so, the user won't get electrocuted
-

Fuse values:

- The fuse value tells you the current needed to blow the fuse
- It must be greater than the normal current, but as close to it as possible, so that the fuse will blow as soon as the current gets too high

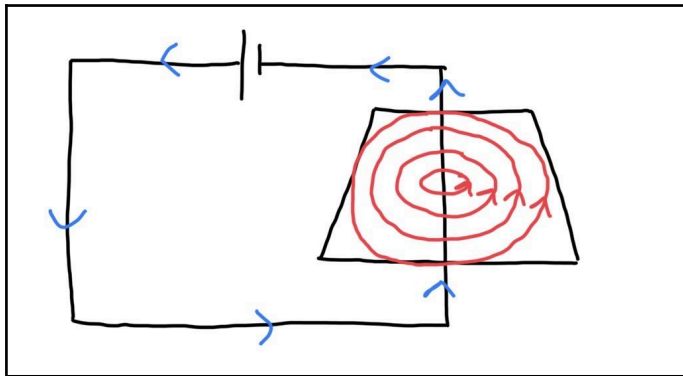
Example: you have 3 fuses: 3A, 5A and 7A.

- If the normal current is 4A, choose the 5A fuse

4.5.3. Magnetic effect of a current

Magnetic field around a conducting wire

- When a current flows through a conducting wire, a magnetic field is induced around the wire

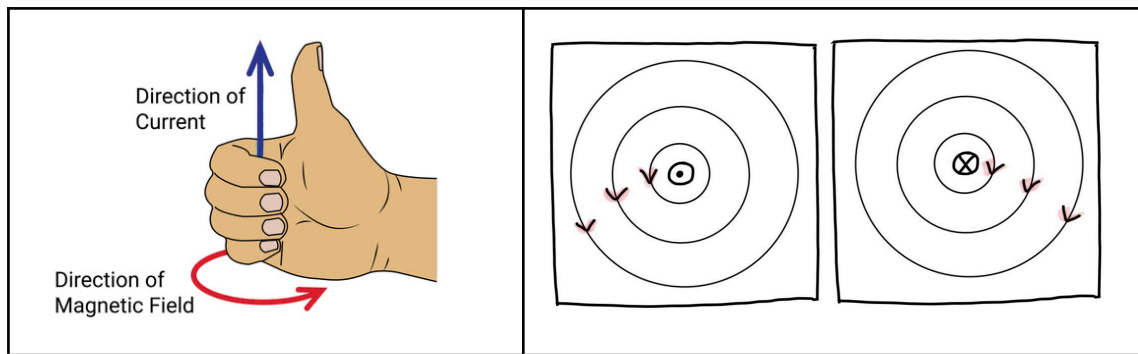


Magnetic field line patterns:

- Look like circles around the wire
- Distance between each circle increases as it gets further from the wire -> This shows that the magnetic field produced is strongest closest to the wire, and gets weaker as distance from wire increases

Right hand grip rule:

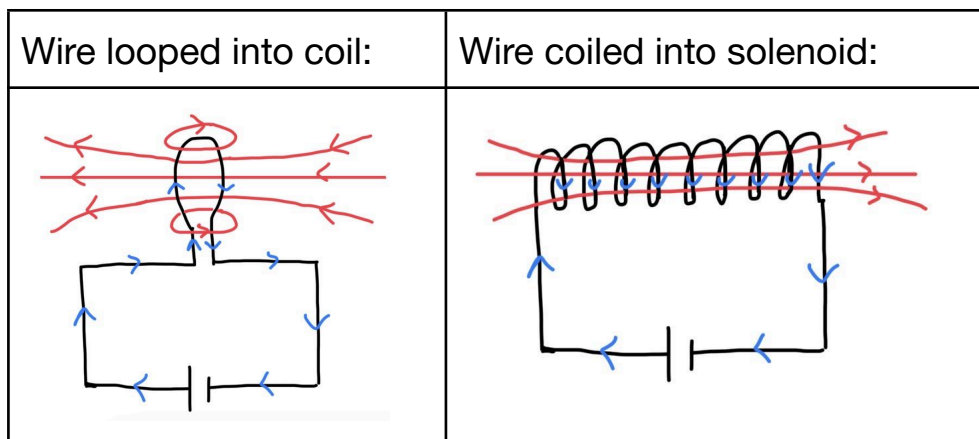
- To determine direction of magnetic field / current



Other things:

- Reversing the current direction will reverse the magnetic field direction
- Increasing the current will increase the magnetic field strength -> so, magnetic field line will be closer together

Magnetic field around a solenoid

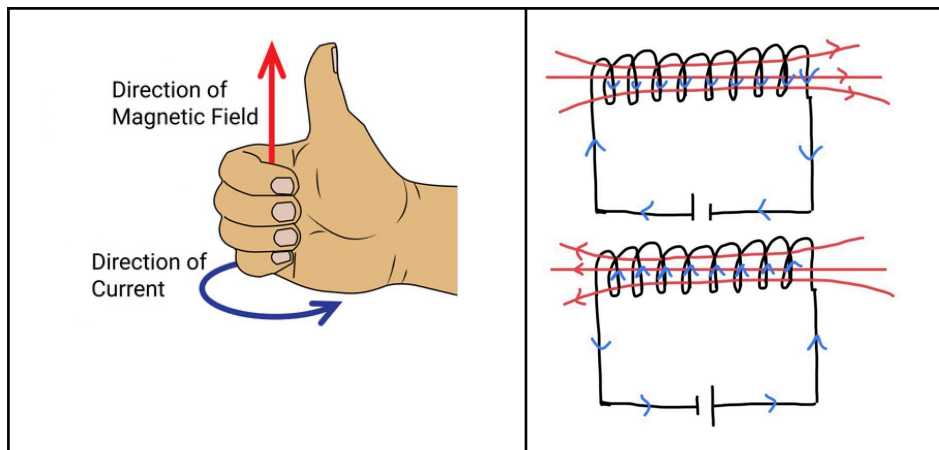


Magnetic field patterns:

- Similar to a bar magnet
- Magnetic field inside coil / solenoid is strong and uniform

Right hand grip rule -> but again, it is switched for coil and solenoid

- To determine direction of magnetic field and current



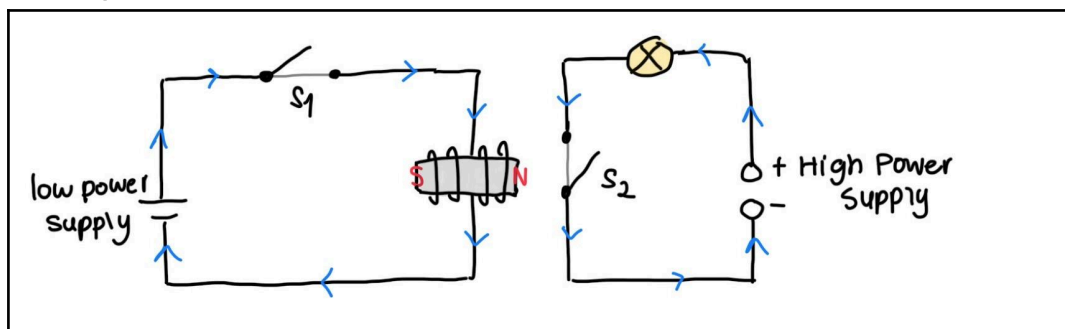
Other things:

- Reversing the current direction will reverse the magnetic field direction
- Increasing the current will increase the magnetic field strength -> so, magnetic field lines are closer together
- Adding more turns on the solenoid will increase magnetic field strength

Electric relay

- A switch operated by an electromagnet
- This uses a small current flowing through one circuit, to turn on a large current in another circuit

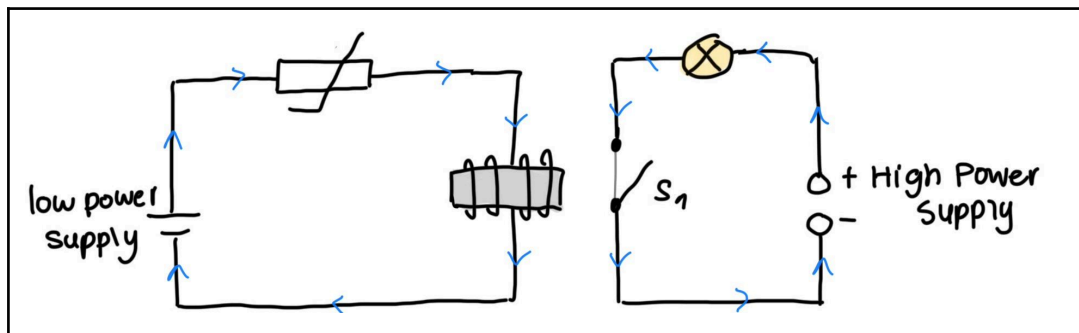
Example 1:



- When switch S1 is closed, a small current flows through this circuit
- The current flowing through the relay coil induces a magnetic field

- This magnetic field attracts switch S2, causing it to close
- This allows a large current to flow in the other circuit, causing the light bulb to turn on

Example 2:

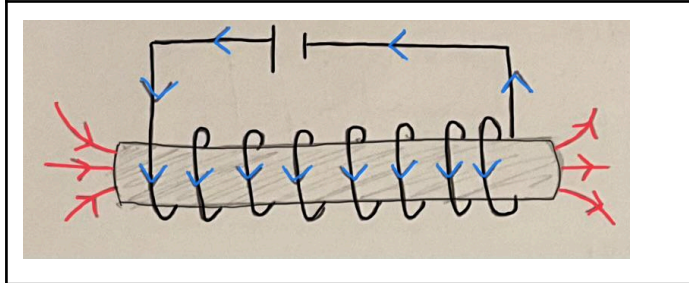


- As temperature increases, resistance of thermistor decreases
- This causes current in this circuit to increase, causing the magnetic field induced around the relay coil to increase
- The stronger magnetic field attracts the switch (S) in the other circuit, causing it to close
- This allows a large current to flow in the other circuit, causing light bulb to turn on

Electromagnet

- A solenoid can be used as an electromagnet, by adding a soft iron core

- Iron core is a soft magnetic material which is easily magnetised and demagnetised, allowing electromagnet to be switch on and off by turning current on and off



- When current flows, a magnetic field forms around the electromagnet
- When it is switched off, the magnetic field disappears

Strength of an electromagnet can be increased by:

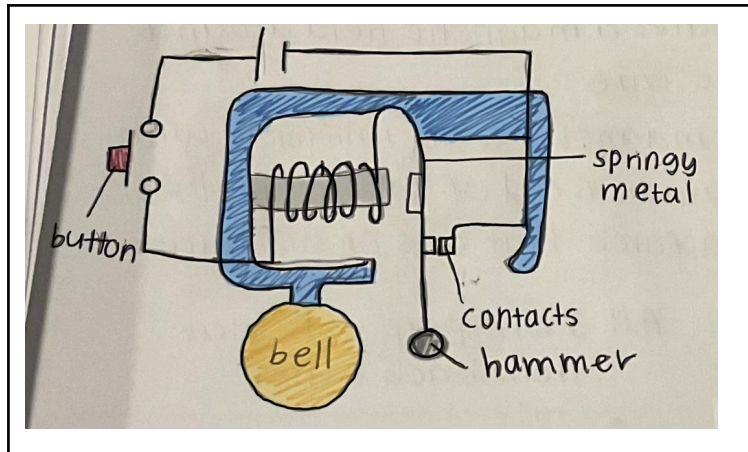
- Increasing current
- Increasing the number of turns on the solenoid
- Adding a soft iron core

Applications of electromagnets:

- Electric relays
- Loudspeakers
- Headphones

Electric Bell:

- Relies on electromagnet to function

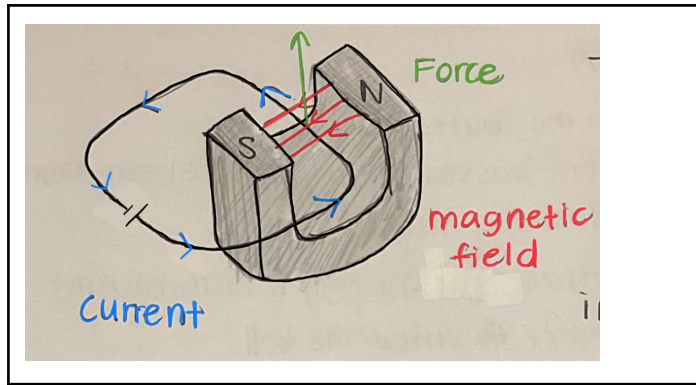


- When the button is pressed, a current passes through the electromagnet, inducing a magnetic field
 - This magnetic field attracts the springy metal, causing the hammer to strike the bell
 - When the hammer strikes the bell, this stops the current by disconnecting the contacts, causing the magnetic field to disappear
 - This causes the springy metal to return to its original position.
-

4.5.4 Force on a current carrying conductor

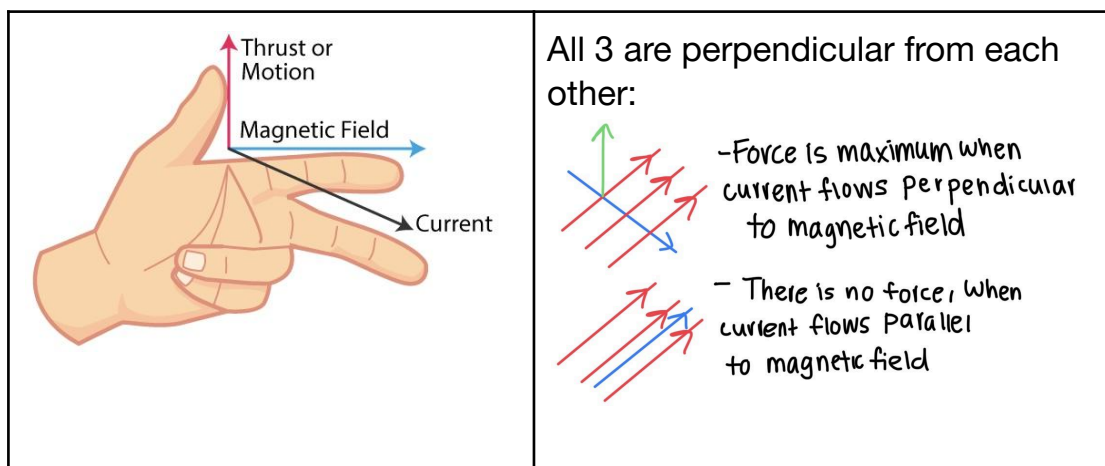
Force on a current carrying conductor

- When a current flows through a wire, it induces a magnetic field around the wire
- This magnetic field interacts with the magnetic field of the magnet, inducing a force that acts on the wire



Flemming left hand rule:

- To determine direction of current, magnetic field of magnet, and force induced (FMI)



Direction of force on a wire can be reversed by:

- Reversing direction of current
- Reversing the magnetic field

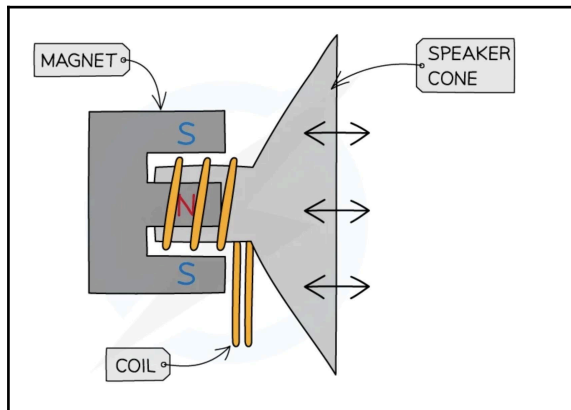
(If both reversed at the same time, direction of force remains the same)

Magnitude or magnetic force can be increased by:

- Increasing the current
- Increasing the magnetic field strength by using strong magnet
- Increasing length of wire that interacts with the magnetic field of magnet, by adding more loops

Loudspeakers and headphones:

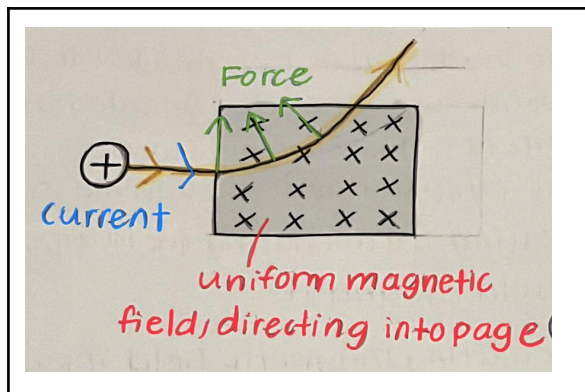
- Converts electrical energy into sound



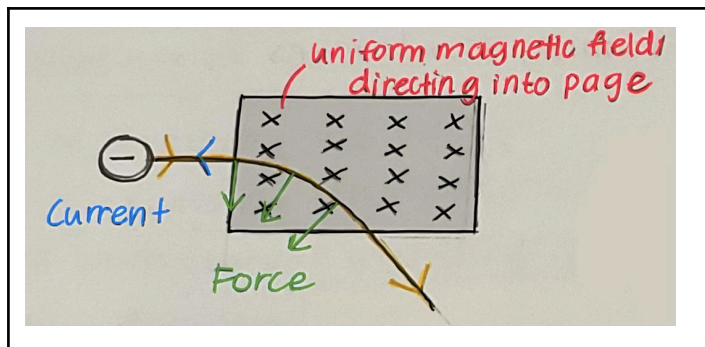
How sound is made in loudspeaker:

- Loudspeaker is connected to an amplifier that provides an alternating current
- This current flows through the coil, creating an alternating magnetic field around it
- The alternating magnetic field interacts with the magnetic field of the magnet, inducing a force on the coil that constantly changes directions
- This causes the coil to oscillate.
- The oscillating coil causes the cone to oscillate, creating compressions and rarefactions, forming sound waves

Force on a charged particle in a uniform magnetic field



- Positively charged particle creates an electric current in the direction of its motion
- The current creates a magnetic field that interacts with the uniform magnetic field, inducing a force that acts on the particle upwards
- Force is always perpendicular to current, so this causes the particle moves in a circular path upwards
- When the particle leaves the uniform magnetic field, it moves in a straight line, because there is no interaction, so no force is induced



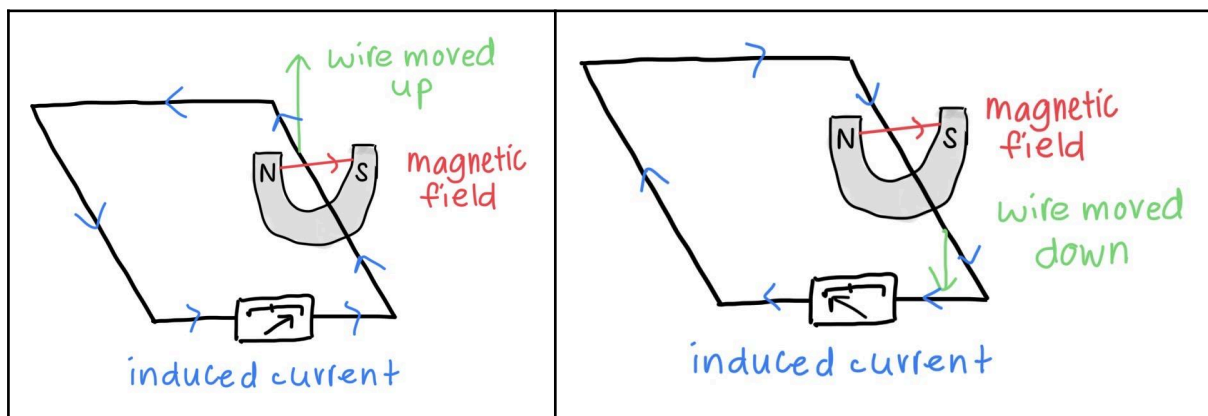
- Negatively charged particle creates an electric current in the opposite direction of its motion
- The current creates a magnetic field that interacts with the uniform magnetic field, inducing a force that acts on the particle downwards
- Force is always perpendicular to current, so this causes the particle to move in a circular path downwards
- When the particle leaves the uniform magnetic field, it moves in a straight line, because there is no interaction, so no force is induced

Note: the movement of the particle (upwards or downwards) can also change depending on the magnetic field (whether into or out of page)

4.5.1 Electromagnetic induction

Electromagnetic induction in a wire

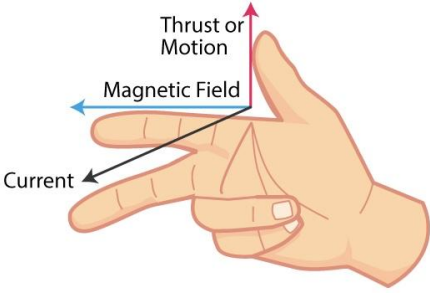
- Electromagnetic induction: the process that emf / voltage / current is induced in the conductor, when it is interacting with a changing magnetic field



- When the wire moves upwards, this cuts the magnetic field lines of the magnet, inducing an emf.
- This causes the needle of the galvanometer to deflect to the right
- When the wire moves downwards, this cuts the magnetic field lines of the magnet, inducing an emf
- This causes the needle of the galvanometer to deflect to the left

Right hand Fleming rule:

- To determine movement of conductor, magnetic field, and induced emf/current

	- When wire moves perpendicular to magnetic field, this induces maximum emf - When wire moves parallel to magnetic field, there is no induced emf
---	--

Direction of induced current can be reversed by:

- Reversing direction of movement of wire
- Reversing direction of magnetic field

(If both are reversed at the same, direction of induced emf remains the same)

Induced current can be increased by:

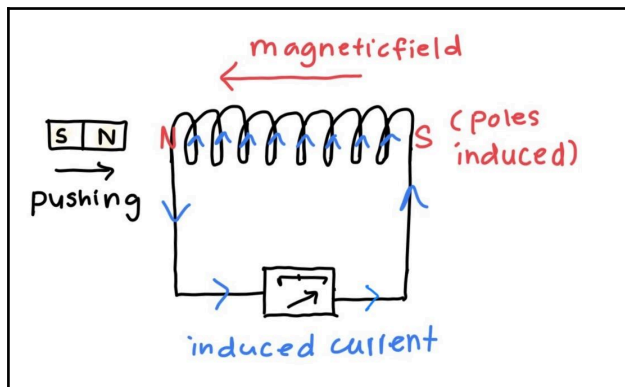
- Increasing magnetic field strength, by using a stronger magnet
- Moving wire faster
- Increasing length of wire that interacts with the magnetic field of the magnet, by adding more loops

Note: when wire and magnet is stationary, no induced current -> because, no magnetic field lines are cut

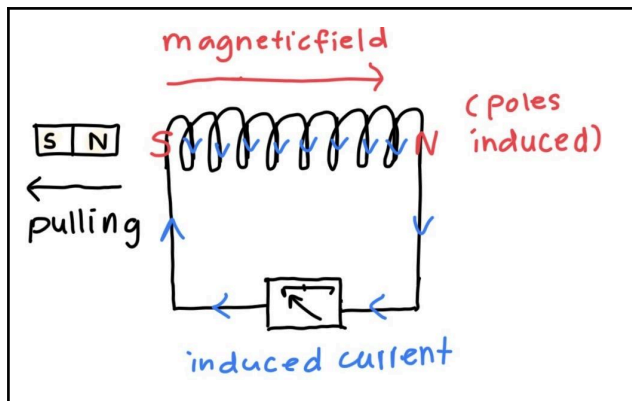
- When induced current increases, there is a larger deflection of the needle.

Electromagnetic induction in a solenoid

- Lenz law - the direction of induced emf always opposes the change causing it
- Cuts coils -> induced emf in solenoid -> creation of magnetic field around solenoid -> opposes change in magnetic field of magnet



- When the magnet is moved into the solenoid, the magnetic field lines of the magnet cuts the coils of the solenoid, inducing an emf and current in the solenoid
- The induced current in the solenoid creates a magnetic field around the solenoid.
- This magnetic field opposes the change in the magnetic field of the magnet.
- This results in a force exerted on the magnet, in the opposite direction of its motion.



- When the magnet is pulled away, out of the solenoid, the magnetic field lines of the magnet cuts the coils of the solenoid, inducing an emf and current in the solenoid.
- The induced current in the solenoid creates a magnetic field around the solenoid.
- This magnetic field opposes the change in the magnetic field of the magnet.
- This results in a force exerted on the magnet, in the opposite direction of its motion.

Other things

Direction of induced current can be reversed by:

- Reversing the motion of the magnet
- Reversing the polarity of the magnet

Induced current can be increased by:

- Increasing magnetic field strength, by using a stronger magnet
- Moving magnet into or out of solenoid faster
- Increasing the number of turns in the solenoid

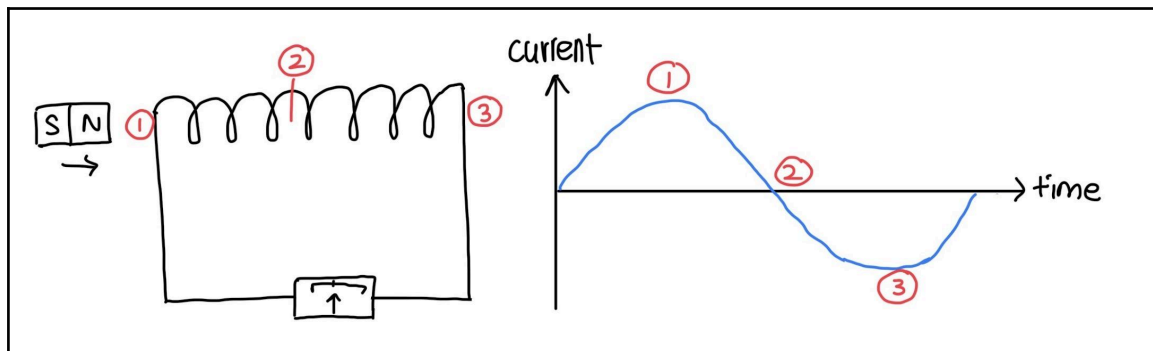
Note: when magnet and solenoid are stationary, no induced current -> because, no magnetic field lines are cut

- When induced current increases, there is a larger deflection of the needle.

Graphs

- This is for when a magnet is continuously going through the solenoid (it is not stationary at any point)
- Again, direction depends on the information given in the question

When magnet is pushed into and out of the solenoid:



1) When magnet enters solenoid:

- Induced current is maximum
- Needle will deflect to one direction
- Magnetic field strength is strongest at its poles

2) At the middle of the solenoid:

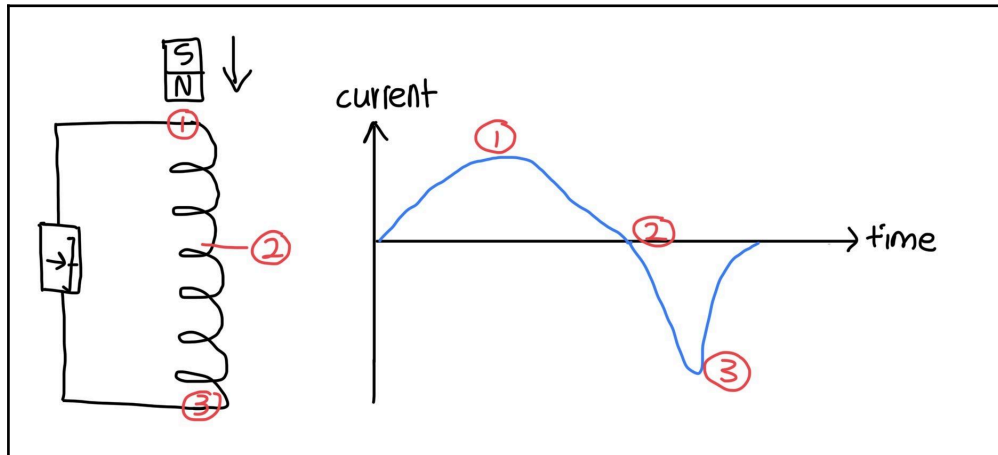
- Induced current is 0
- Induced currents on both sides of the solenoid are equal and opposite in direction, cancelling each other out

3) When magnet leaves the solenoid:

- Induced current is maximum, but in the opposite direction
- Needle will deflect to the opposite direction
- Magnetic field strength is strongest at its poles

- Why reversed as it enters and leaves - change in the magnetic field of the magnet is opposite as it enters and exits, which causes magnetic field of the solenoid and current in the solenoid to reverse.

When magnet is dropped through the solenoid:



1) When magnet enters solenoid:

- Induced current is maximum
- Needle will deflect to one direction
- Magnetic field strength is strongest at its poles

2) At the middle of the solenoid:

- Induced current is 0
- Induced current in both sides of the solenoid are equal and opposite in direction, cancelling each other out

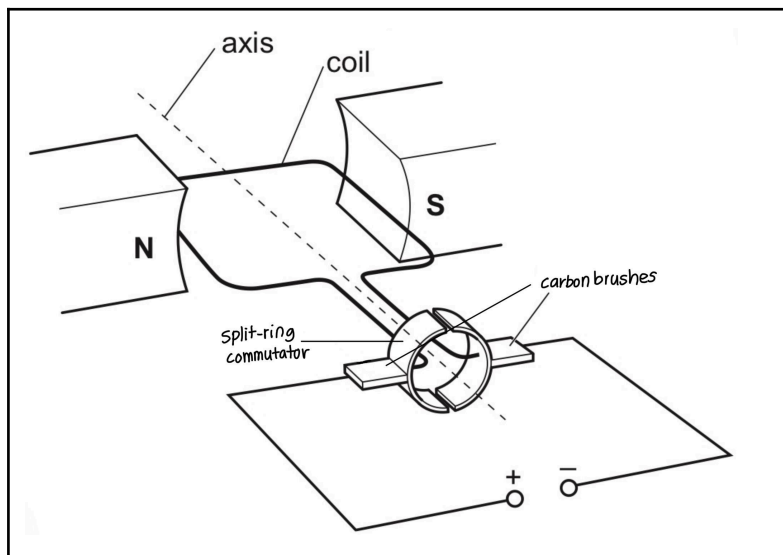
3) When magnet leaves the solenoid:

- Induced current is at a higher maximum, and now in the opposite direction
 - Magnetic field strength is strongest at its poles
 - At a higher maximum, because speed of magnet when leaving is more than when entering, due to gravity.
- Why reversed as it enters and leaves - change in the magnetic field of the magnet is opposite as it enters and exits, which causes

magnetic field of the solenoid and current in the solenoid to reverse.

4.5.5 The DC Motor

- DC motor changes electrical energy into kinetic energy



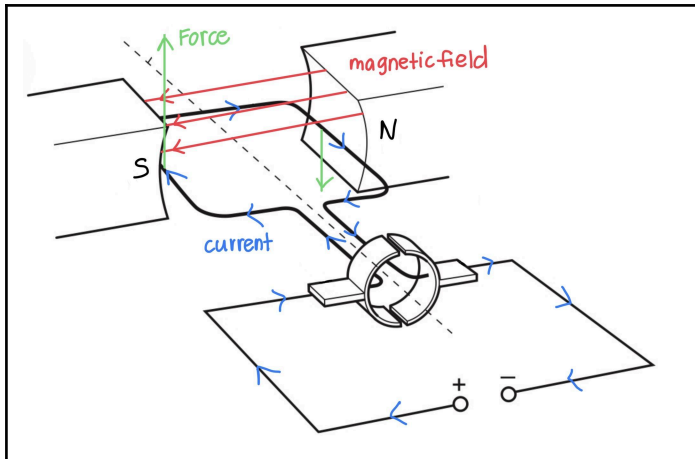
Split-ring commutator: keeps coil rotating in the same direction, by reversing the current every half a revolution

Carbon brushes: ensures continuous contact for electrical connection between coil and external circuit

How it works

- Uses Left hand flemming rule

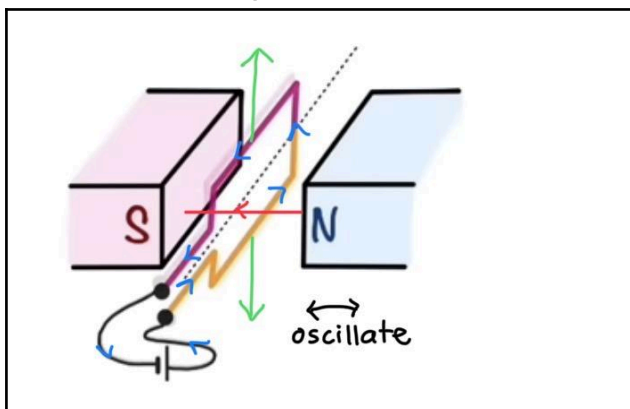
In the horizontal position:



- When current flows in the coil, the current creates a magnetic field around the coil
- This magnetic field interacts with the magnetic field of the permanent magnet, inducing forces that act on both sides of the coil
- The force acts upwards on the left side of the coil
- The force acts downwards on the right side of the coil
- This causes the coil to rotate clockwise, and it will continue to rotate until it is in the vertical position.

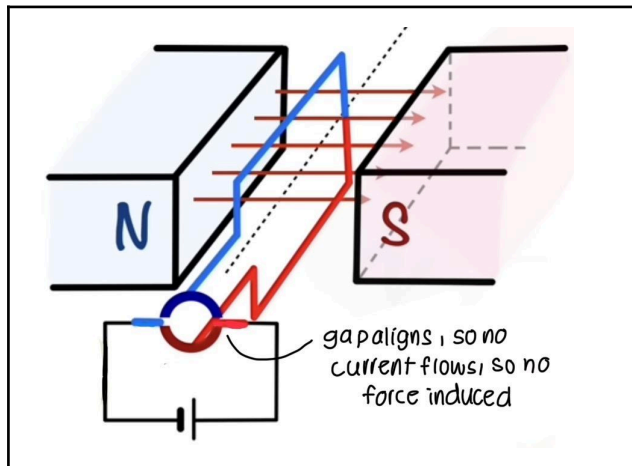
In the vertical position

If without the split ring:



- The forces on each side of the coil act upwards and downwards, resulting in no net turning effect
- The momentum (inertia) of the coil keeps it turning until the force induced causes the coil to rotate in the opposite direction, anti-clockwise
- This keeps taking place, causing the rotation to constantly change direction
- This causes the coil to oscillate with a small angle

If with the split ring:



- The gap in the split ring aligns with the carbon brushes
- No current flows, causing no forces to be induced and acting on the coil
- Momentum (inertia) of the coil keeps it turning until the split rings connect to the carbon brushes again, but now on the other side
- This reverses the current in the coil, which keeps the coil rotating in the same direction (clockwise)

The speed at which the coil rotates can be increased by increasing the magnitude of the force induced, by:

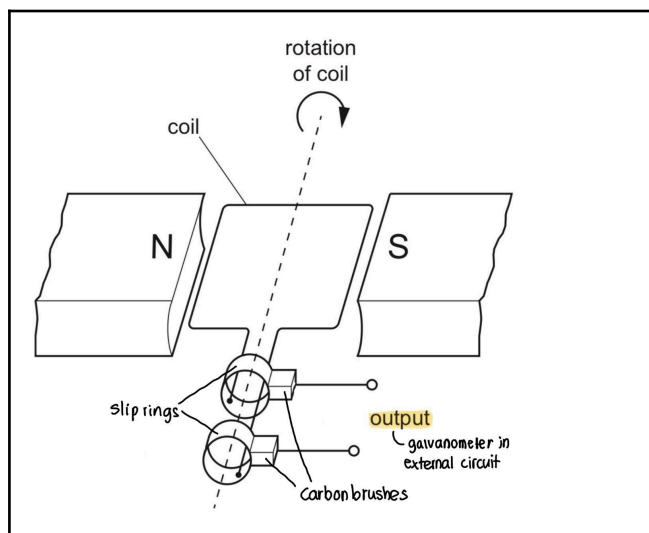
- Increasing the current
- Increasing the magnetic field strength by using a magnetic
- Increasing the number of turns in the coil

The direction of rotation of coil in the DC motor can be reversed by reversing the direction of the forces acting on the wire, by:

- Reversing the current
- Reversing the direction of the magnetic field by reversing the poles of the magnet

4.5.2 The AC Generator

- AC generator converts kinetic energy into electrical energy



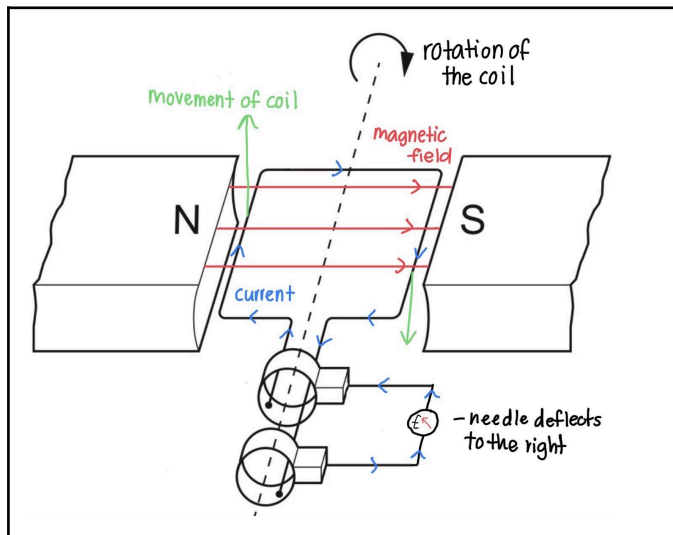
Slip rings: ensures coil is able to rotate without getting tangled

Carbon brushes: ensures continuous contact for electrical connection between coil and external circuit

How it works

- Uses right hand flemming rule

General:

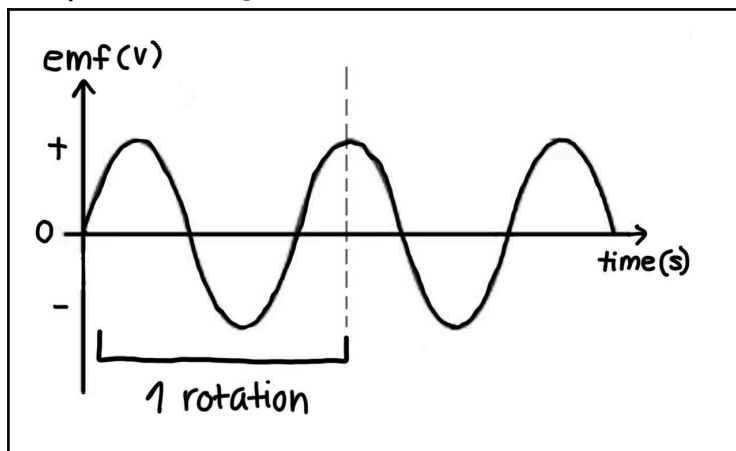


- Turning the handle causes the coil to rotate
- Both sides of the coil cut magnetic field lines, inducing an emf. This induces a current to flow in the coil

Motion of an AC generator:

	- A maximum emf is induced when movement of coil is perpendicular to magnetic field - The greatest number of magnetic field lines are cut by the coil
	- No emf is induced when movement of coil is parallel to magnetic field - No magnetic field lines are cut by the coil

Graph for AC generator:



Maximum emf:

- Movement of coil is perpendicular to magnetic field
- Most magnetic field lines are cut

Induced emf then decreases:

- The angle between movement of coil and magnetic field lines decreases
- Magnetic field lines being cut decreases

Induced emf is 0:

- Movement of coil is parallel to magnetic field
- No magnetic field lines are cut

Then increases to maximum emf again, but in opposite direction

Then decreases to 0, then increases to maximum emf in opposite direction

(Then the whole process repeats)

Induced emf can be increased by:

- Turning coil faster
- Increasing magnetic field strength by using stronger magnet
- Increasing number of turns on the coil

When the coil rotates faster:

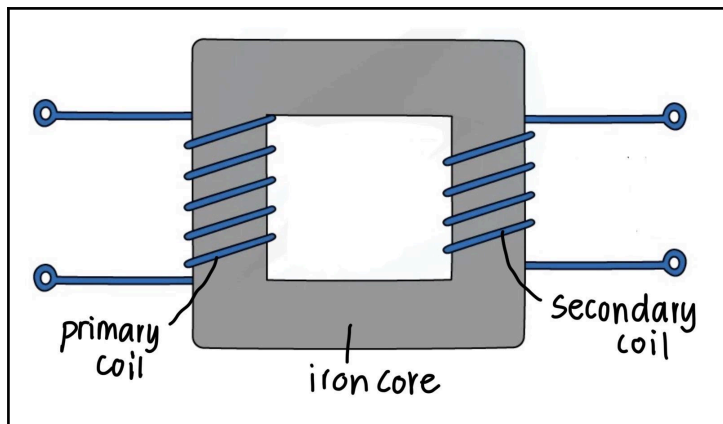
- Maximum induced emf increases
- Time period of alternating emf decreases
- Frequency of alternating emf increases

4.5.6 Transformers

Transformers

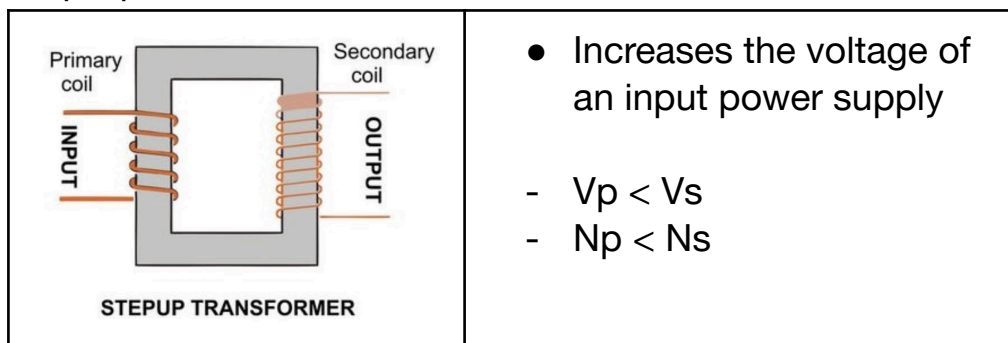
- Electrical device that is used to increase or decrease the voltage of an AC current
- An iron core is used -> this is because it is a soft magnetic material that is easily magnetised and demagnetised
- It is used to allow changing magnetic field to pass through it from the primary to the secondary coil

How they work:

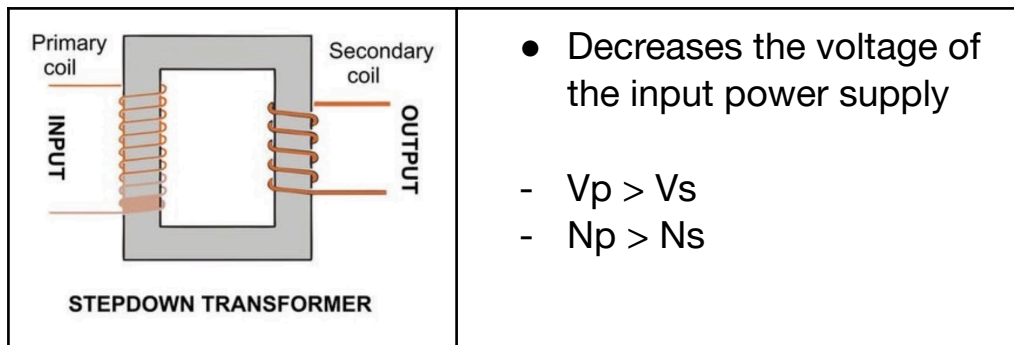


- When alternating current flows through the primary coil, this creates a changing magnetic field around it
- The iron core is easily magnetised, and the changing magnetic field passes through the iron core to the secondary coil
- The changing magnetic field cuts the secondary coil, inducing an alternating current in the secondary coil

Step-up transformer:



Step-down transformer:



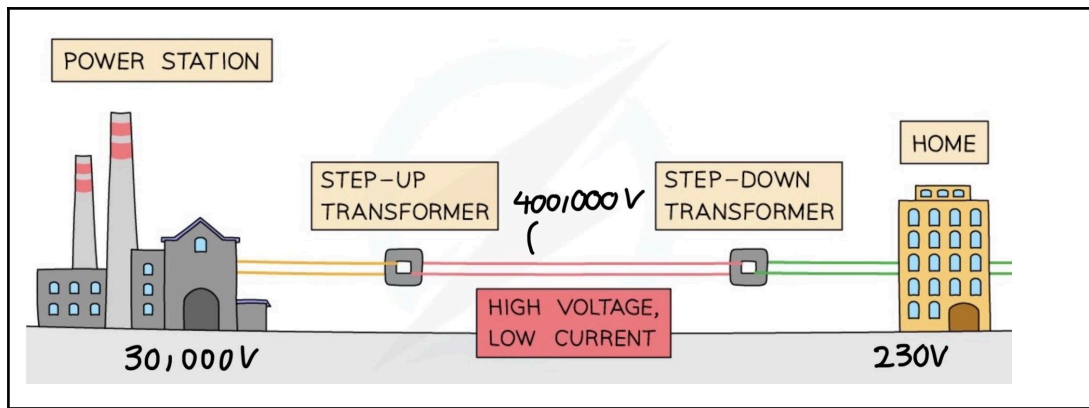
Assuming all Magnetic field lines pass through both coils, and there is no energy loss due to heating effects:

$$\frac{\text{voltage (primary)} \quad V_p}{\text{voltage (secondary)} \quad V_s} = \frac{N_p \text{ turns (primary)}}{N_s \text{ turns (secondary)}}$$

$$\underbrace{V_p I_p}_{\text{Primary}} = \underbrace{V_s I_s}_{\text{Secondary}}$$

National Grid

- Networks of wires and cables carry electrical energy from power stations, to consumers such as factories and homes
- Transformers are used to increase the voltage, so that there is a smaller current in the cables (and less power loss)



High-voltage transmission:

- Electricity is transmitted through cables at a lower current, to prevent power loss
- When current flows in a wire, there is heating effect due to resistance
- Energy is lost as heat to the surroundings
- So, the lower the current, the more efficient the energy transfer. This reduces power loss.

Evidence -> Electrical Power

$$P = IV$$

power (w) current (A) voltage (V)

- By Increasing voltage, there is a lower current, so less heat is produced and less energy is lost to the surroundings as heat

Advantages of high-voltage transmission:

- Reduces power loss, increasing efficiency of energy transfer
- Lower currents means thinner cables can be used, which is cheaper

Calculating Power loss:

$$P = I^2 R$$

power (W) current (A) resistance (Ω)